

SenNet Modular

*Modular Metering System
Installation Manual*



Revision 1.04

SenNet Modular

Modular Metering System - Description#

1. Description

The **SenNet Modular** electricity metering System simultaneously monitors up to 20 three-phase loads or up to 60 single-phase loads (or a combination of both load types). The system integrates load measurement I/O functions logging and communications in a single, flexible unit which can be tailored to suit a variety of energy management installations.

SenNet Modular systems are supplied as a combination of the following modular components:

- **1 Master Display Unit**
- **1-10 Option Modules**
 - Sub-Metering Modules (e.g. MC352*)
 - I/O Modules (eg Pulse Output Module(s) PLS12)
- **1 Communications Module (e.g. Modbus RS485 or Modbus TCP/IP)**

* At least 1 sub metering module is always fitted as and END MODULE.

1.1. Master Display Unit

The Master Display Unit provides a user interface and local display of metered parameters on a graphic LCD and can optionally log up to 200 days of energy readings from the Sub-Metering Modules.

The Master Display Unit also acts as a power supply and voltage measurement input for all the Sub-Metering Modules, which may be attached to it. This single voltage input point makes wiring much simpler and safer by removing the need for distributed voltage connections.

1.1.1. Master Display Unit Supply

The Master Display unit is available in two versions as follows:

- | | |
|-----------|---|
| Version 1 | Isolated data to SenNet Modular communications modules.
Isolated, 5Vdc unregulated power supply 0.5Watts max. |
| Version 2 | Isolated data to SenNet Modular communications modules.
Isolated, 5Vdc regulated power supply 2Watts max. |



The installation guide supplied with each optional communications module provides details of the minimum dc power supply required for correct operation.

1.2. Dual Sub-Metering Module – MC352

The Electricity Metering Module **MC352** contains two complete 3-phase electricity meters, each of which may be optionally configured to monitor 3 single-phase loads.

Each electricity meter accurately measures a wide range of power and energy parameters using a range of current input devices selected to suit loads with nominal inputs from 5 to 800 Amps. These specially designed transducers each have a nominal output of 0.33V, are safety-isolated and internally protected against high open-circuit voltages at the output.

Split core current input devices can be fitted to existing power cables where it is inconvenient to remove one end of the cable for connection. Miniature ring type devices are also available for lower currents (up to 60A) providing a lower cost solution, with improved accuracy, where it is possible to slide these over one end of a power cable. Dual Sub-Metering Modules are configured using the Master Display Unit LCD/keypad interface or via the external communications network.

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Modular Metering System - Assembly

1.3. *Communications Module*

The Communications Module provides a connection point to external systems such as building energy management, billing data collection, SCADA etc. A specific Communications Module may be selected, such as the Modbus RTU - RS485 Module (RTU485), to suit a range of external systems.

The Communications Module provides external access to:

- **Master Display Unit Configuration**
- **Master Display Unit Logged Data (Optional)**
- **Sub-Metering Module Instantaneous Meter Readings and Energy Registers**
- **Sub-Metering Module Configuration**
- **Additional Module Data and Configuration**

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Modular Metering System - Safety#

2. Safety

This manual gives details of safe installation of **SenNet Modular** electricity metering systems. Safety may be impaired if the instructions are not followed or the system is used in a manner not specified by the manufacturer. Labels give details of equipment ratings for safe operation. Take time to examine all labels before commencing installation. Safety symbols on the meter have specific meanings.



Safety may be impaired if the instructions are not followed or the meter is used in a manner not specified by the manufacturer.



Contains no user serviceable parts. Field wiring and commissioning should only be carried out by qualified personnel, in compliance with applicable national regulations.
e.g. National Electrical Code (NEC) for US; Canadian Electrical Code for Canada

2.1. Maintenance

The equipment should be maintained in good working order. Damaged equipment must be returned to the manufacturer (or his authorised agent) for repair. The meter may be cleaned by wiping lightly with a soft cloth. No solvents or cleaning agents should be used. All inputs and supplies must be isolated before cleaning any part of the equipment.

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Modular Metering System - Installation#

3. Installation

3.1. Intended Use

A **SenNet Modular** System simultaneously monitors up to 20 three-phase loads or up to 60 single-phase loads or a combination of both load types. The system is designed for connection to the following power networks:

- **3-Phase 3 or 4 Wire: Nominal 230V Phase to Neutral (400V Phase to Phase)**
- **Single Phase 2 Wire: Nominal 230V Phase to Neutral.**
- **Two Phase 3-Wire: Nominal 120V Phase to Neutral (240V Phase to Phase)**
- **Optional - 3-Phase 3 or 4 Wire: Nominal 277V Phase to Neutral (480V Phase to Phase)**

The **SenNet Modular** should be mounted close to the power system distribution point to minimise the length of voltage and current input connections. Ensure that the voltage cable clamps are securely tightened and all terminal covers are securely fitted before powering up the system.



The meter is designed for connection in an installation with overvoltage category III. This category (CAT III) is for equipment installed at or near the origin of the electrical supply to the building, between the building entrance and the main distribution board.

3.2. Mounting

The **SenNet Modular** is designed to be mounted on a 35mm symmetrical ("Top-Hat") DIN rail. The minimum length of DIN Rail required depends on the number of standard modules fitted as follows:

<i>Comms Module</i>	<i>Master Display Unit</i>	<i>Standard Modules</i>	<i>Right Hand End Module</i>	<i>Overall Length</i>
29mm	100mm	0 = 0mm	29mm	158mm
29mm	100mm	1 = 29mm	29mm	187mm
29mm	100mm	2 = 58mm	29mm	216mm
29mm	100mm	3 = 87mm	29mm	245mm
29mm	100mm	4 = 116mm	29mm	274mm
29mm	100mm	5 = 145mm	29mm	303mm
29mm	100mm	6 = 174mm	29mm	332mm
29mm	100mm	7 = 203 mm	29mm	361mm
29mm	100mm	8 = 232mm	29mm	390mm
29mm	100mm	9 = 261mm	29mm	419mm

To clip the **SenNet Modular** on the DIN rail first hook onto the top of the rail then push the bottom of the unit till all the sprung clips click into place. For wall mounting the DIN rail should be first secured to the wall and the **SenNet Modular** system clipped on afterwards.

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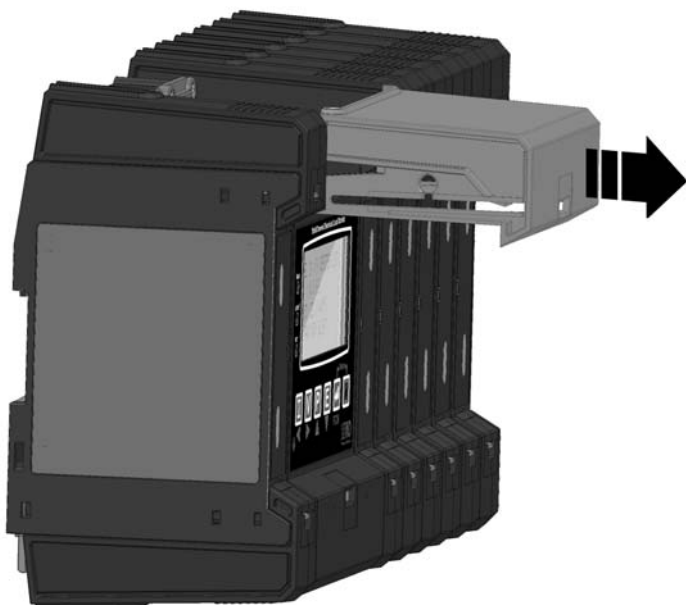
Modular Metering System - Installation

3.2.1. Access to Master Display Unit

The auxiliary mains supply and voltage input terminals are protected by a sliding terminal cover on the Master Display unit. To gain access to these terminals:



1. Insert a flat screwdriver into the slot at the front of the terminal cover.



2. Slide the terminal cover forward to its full extent.
3. Hinge the cover down in front of the display.

NOTE: The screwdriver slot may be covered with a tamper evident label (20mm x 18mm) to detect unauthorized access.

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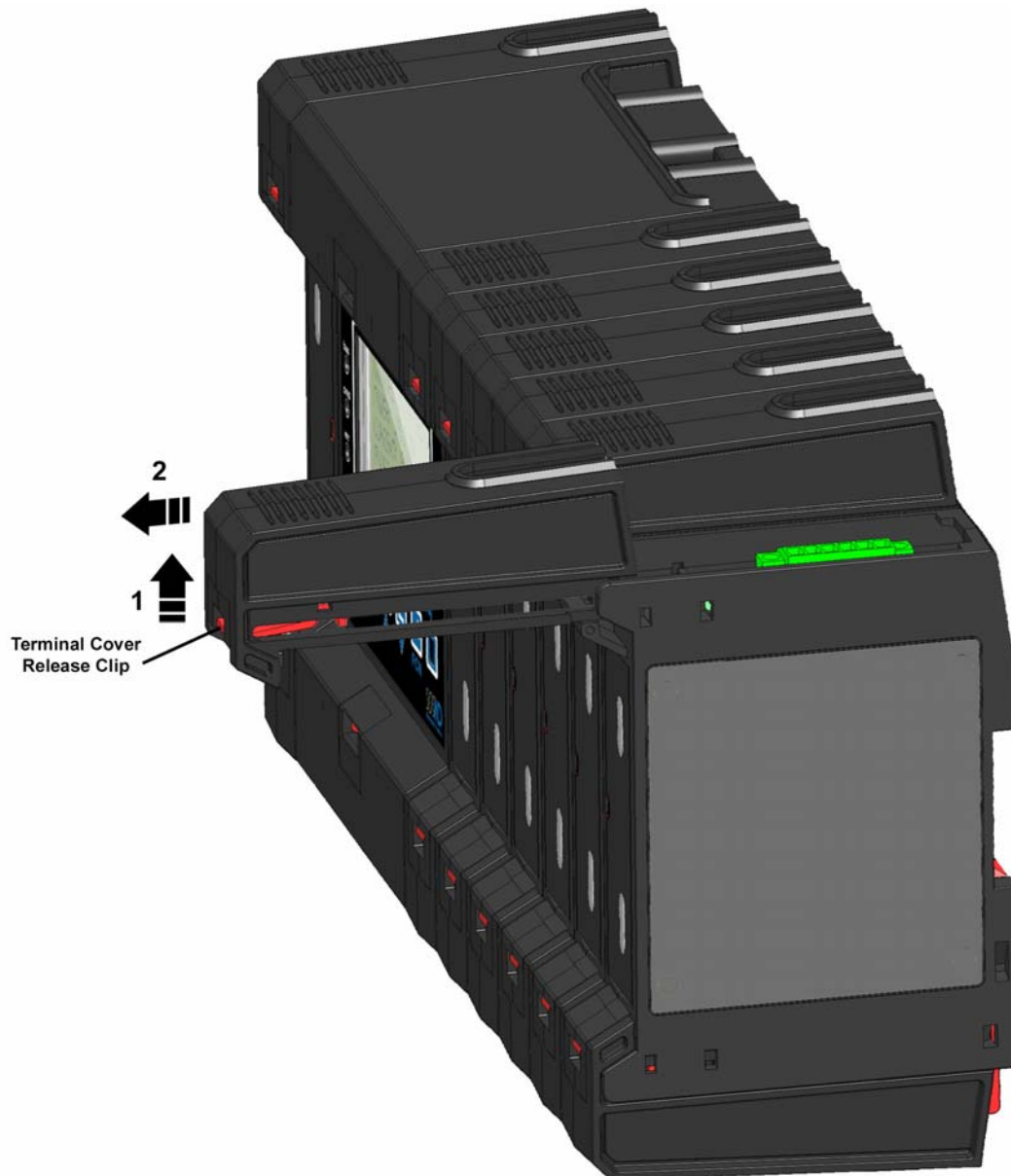
Modular Metering System - Installation

3.2.2. Accessing Module Wiring Terminals

Module wiring terminals are protected by terminal covers as shown in the diagram below. The terminal covers are designed to slide towards the front of the module and are captive so they may not be removed and misplaced.

To access the terminals:

1. Push up the terminal cover release clip using a tool.
2. Slide the terminal cover forward to its full extent
3. Flip the terminal cover to sit in front of the module while accessing the wiring terminals.



NOTE: The release clip may be covered with a tamper evident label (20mm x 18mm) to detect unauthorized access.

4. Schematics

Each ½ of an **MC352** Metering Module acts as an independent meter and can be configured during commissioning to monitor a three-phase load or three single-phase loads.

4.1. Voltage Circuits Safety



A suitably located and easily reached switch or circuit breaker must be included as part of the installation. This could, for example, be a safety-interlocking device on the door/front panel of the electrical enclosure. This switch/circuit breaker must be marked as the disconnecting device for the equipment and must comply with the relevant requirements of IEC 60947-1 and IEC 60947-3.



Disconnect / Isolate all supplies before commencing installation.



To maintain proper insulation from the mains supply, the measurement neutral should only be connected in power networks where the system neutral is protectively earthed



Voltage cables must be rated for safe use in the electrical enclosure which houses the meter (e.g. UL1015) and must meet the following minimum specification:
Temperature: 105°C; Insulation 600Vac.

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Modular Metering System - Schematics

4.2. Auxiliary Supply

The **MC352** requires an auxiliary mains power supply to power the internal electronic circuits. The auxiliary supply circuits must meet or exceed the following specification:

Terminations:

Cable: UL-1015; 600Vac; 105°C. 30-12AWG, Stripped 5.5 to 6.5mm
Torque: 0.5Nm (4.4in lb)

Fuses (US/Canada)

Rated Voltage	Type	Rupture In (A)	Standards
≥240Vac	Time-Delay	1.0A	UL 248 (US) C22.2 No. 248 (CAN)

Fuses (Other Countries)

Rated Voltage	Type	Rupture In (A)	Standards
≥240Vac	Time-Delay	1.0A	IEC 60269-2

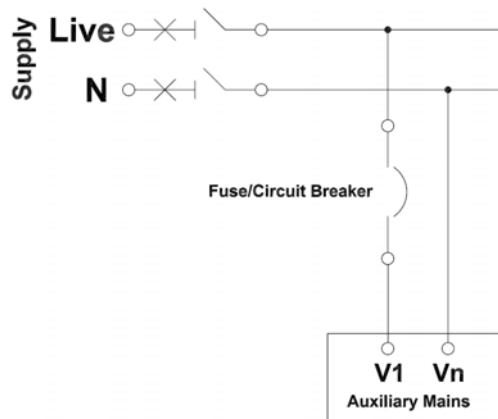


Figure 4-1 Auxiliary Mains Connection

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Modular Metering System - Schematics

4.3. Voltage Measurement Inputs

Voltage measurement connections are common for all the slave meters and are made at the Master Display Unit. The voltage inputs are attenuated before being passed on to all metering modules. The voltage input and auxiliary supply circuits must meet or exceed the following specification:

Terminations:

Cable: UL-1015; 600Vac; 105°C. 30-12AWG, Stripped 5.5 to 6.5mm
Torque: 0.5Nm (4.4in lb)

Fuses (US/Canada)

Rated Voltage	Type	Rupture In (A)	Standards
≥500Vac	Time-Delay	1.0A	UL 248 (US) C22.2 No. 248 (CAN)

Fuses (Other Countries)

Rated Voltage	Type	Rupture In (A)	Standards
≥500Vac	Time-Delay	1.0A	IEC 60269-2

The auxiliary mains supply may be connected to any one of the measurement phases and system neutral as shown below (Phase 1 and Neutral shown).

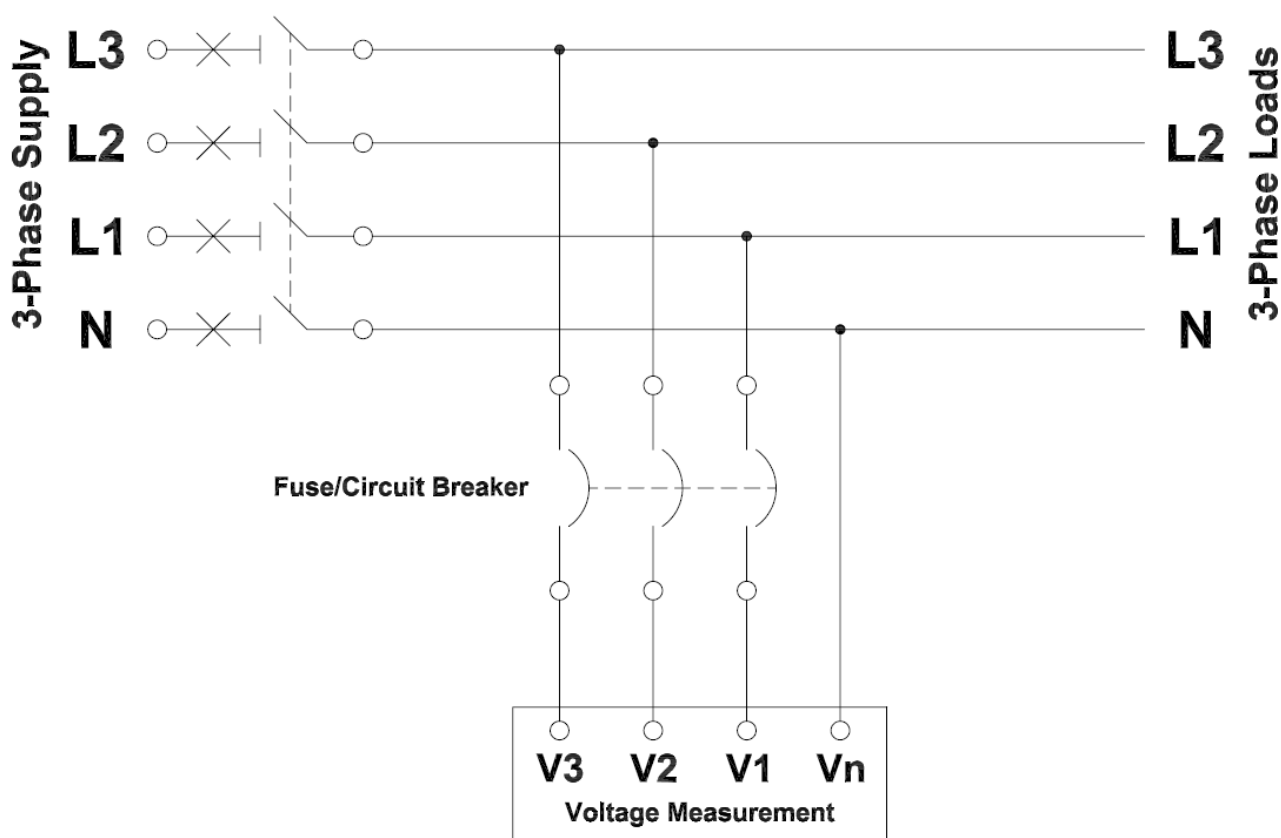


Figure 4-2 Measurement Voltage Inputs

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Modular Metering System - Schematics

4.4. Current Inputs MC352

Each phase current input of the **MC352** Sub-Metering Module is associated with one of three phase voltage inputs, which are supplied as an attenuated signal from the Master Display Unit. It is essential that current transducers are mounted on the correct associated voltage phase for accurate measurement.

To ensure accurate operation, the Sub Metering Module **MC352** *should* only be used with Current Transducers supplied by the manufacturer. These provide an isolated 0.33Vac signal to the current inputs when measuring a nominal primary current.



Only current transducers which meet the manufacturer's specifications should be used.

Minimum Current Transducer Specification:

Input Current Range:	0 to 1.2 I _n (I _n = nominal rated current in amps)
Output Voltage:	0.33Vac at I _n
Insulation:	600Vac (Core to secondary conductors)
Internal Isolation:	2.21kV
Cable:	Operating Temperature, 105°C Insulated 600Vac Safety Compliant (e.g. UL1015)

If the current transducer secondary cables require extending, care must be taken to avoid pickup of electrical interference. With suitable low capacitance screened cables, the cable can be extended to 100m or more.



Current Transducer (CT) connections are not galvanically isolated from the voltage inputs and must therefore not be accessible to the operator after installation. Installed CT cables and any extensions to these, must not be accessible to the operator.



*Extensions to the supplied current transducer cables must ensure all connections remain inaccessible to the operator after installation.
All cables and connections must meet the minimum specifications provided.*

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Modular Metering System - Schematics

4.5. Measuring 3 Phase Loads

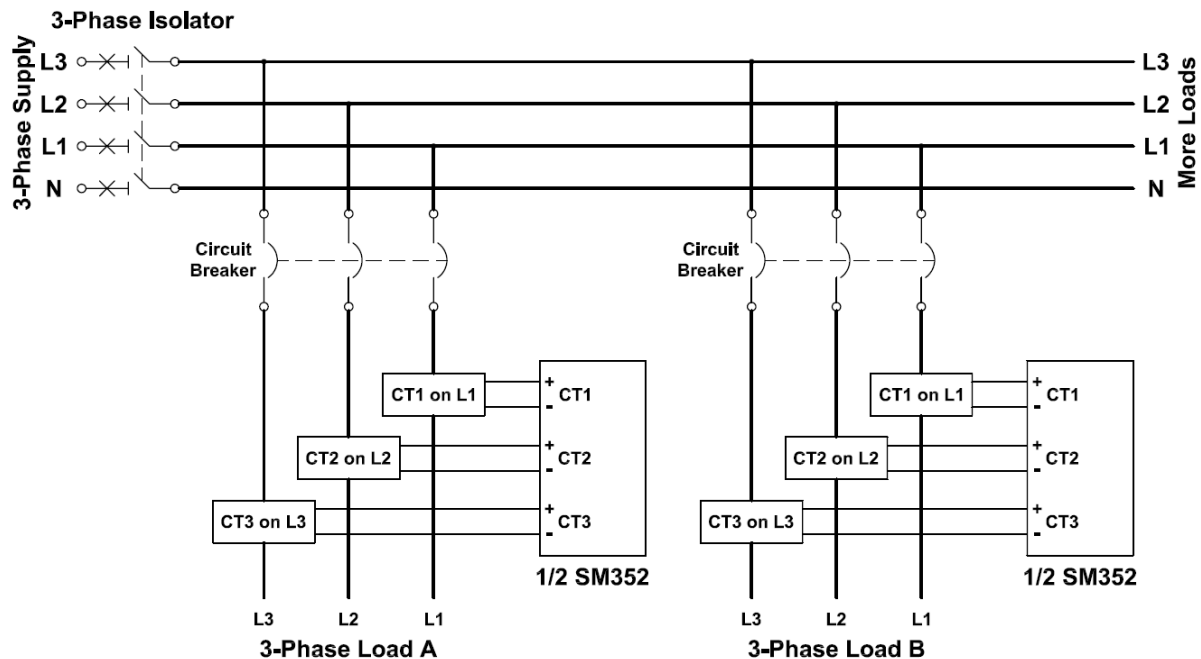


Figure 4-3 Measuring 2 x 3-Phase 3 or 4 Wire Loads on a MC352

4.6. Measuring Single Phase Loads

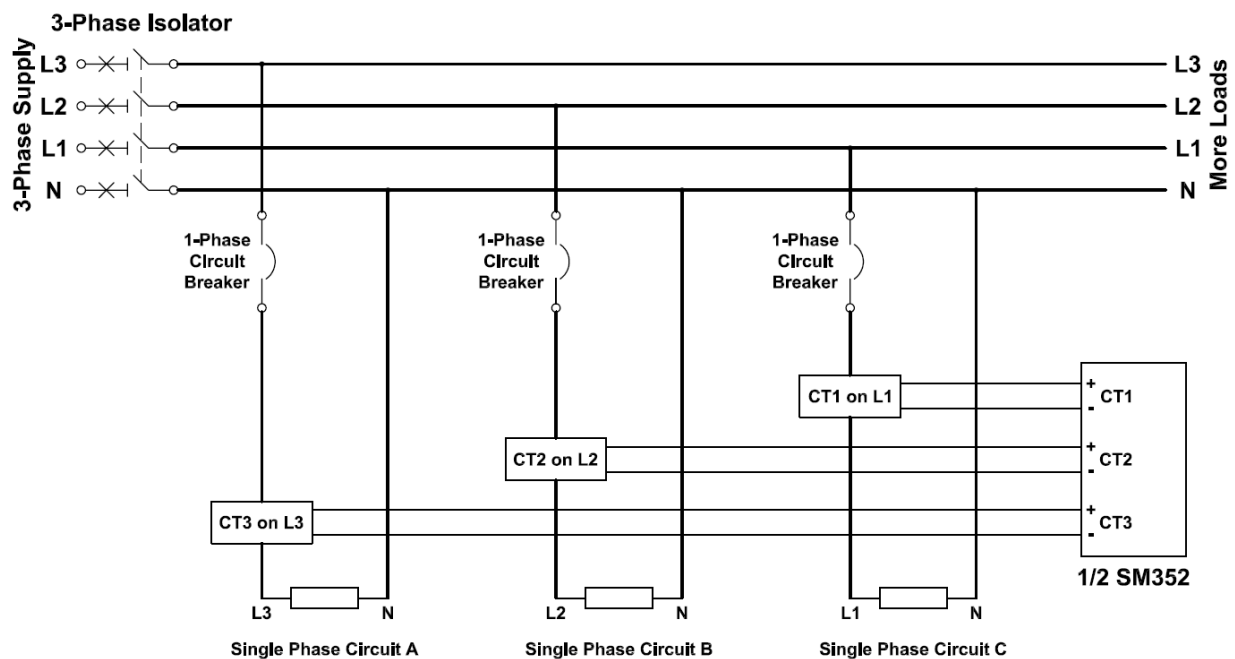


Figure 4-4 Measuring 3 x Single Phase Loads Using 1/2 MC352

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Modular Metering System - Configuration

5. Power Up/Configuration

5.1. Powering up a *SenNet Modular* System

Before supplying power to the *SenNet Modular* system check all wiring, ensure the unit is securely mounted to a stable surface and clean up all debris, scraps of wire etc.

When power is applied to a *SenNet Modular* a system hardware check is carried out to determine which modules are connected and how these are configured. The power up screen displays the progress of this system configuration check.



Symbol	Meaning
	Module Position Empty
	Module not recognised by the system
	Recognised Module

Other information on the Power up screen, such as software version, may be required when contacting the manufacturer for technical support.

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Modular Metering System - Configuration

5.2. Configuring a New Modular Metering System

All new **SenNet Modular** systems are factory configured and calibrated with at least the following components:

1. A Master Display unit
2. At least 1 Metering Module (eg **MC352** dual meter).

Default settings are provided which can be user configured during commissioning to suit the site to be monitored and host communication requirements. Default settings are as follows:

Setting	Default Setting
System - Name	"SenNet Modular"
System – Volts Demand Period	30 minutes
System – Amps Demand Period	30 minutes
System – Power Demand Period	30 minutes
System – Logging (optional)	Disabled
System – RS485 (optional)	Modbus ID = 1 Baud = 9600 Parity = None
Meters – Load Name	"< Meter N >"
Meters – Communications	Load Modbus IDs = 2,3,4 From left to right, top to bottom
Meters – CT Type	Split CT 150A-0.33V
Meters – Load Type	3-Phase
Meters-Autorotate	Enabled

On power up, **SenNet Modular** systems recognise and accept their factory configured modules. Changes to the system hardware configuration may be made in the field such as adding/removing modules providing the original minimum component hardware configuration (above) is maintained.

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Modular Metering System - Configuration

5.3. Configuring a Modified SenNet Modular System

Hardware changes made to a **SenNet Modular** system such as adding/removing modules are detected during the power up and may require action by the user to confirm these changes before the system accepts the new configuration.

5.3.1. After Adding New Modules

If one or more new modules are detected during power up checks, the following screen is displayed:



The detected modules are recognised and shown in their position and the user is requested to accept these changes.

Use the   keys to select YES or NO and the  key to confirm the selection.

NOTE: The “Configuration Changed” screen times out after approximately 20 seconds.

5.3.1.1. Selecting NO (default)

If **NO** is selected (or the screen is allowed to timeout with **YES** or **NO** selected):

- The system continues to operate unchanged with previously configured and accepted modules.
- New modules are skipped when selecting loads in normal operation.
- Existing communications IDs (e.g. Modbus) remain unchanged regardless of physical position in the **SenNet Modular** system.
- New modules are not available on the communications (e.g. Modbus) system.

5.3.1.2. Selecting YES

If **YES** is selected.

- New modules are added to the **SenNet Modular** system and made available for display.
- Existing communications IDs (e.g. Modbus) remain unchanged regardless of physical position in the **SenNet Modular** system.
- New modules are available on the communications (e.g. Modbus) system with consecutive IDs that start at the previous largest ID+1.

NOTE: In order for the data logger (optional) to record data from new module(s), the user must STOP the current LOGGING SESSION and start a new one.

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Modular Metering System - Configuration

5.3.2. After Removing Modules

If power up checks determine that one or more modules have been removed, the following screen is displayed:



NOTE: The “Configuration Changed” screen times out after approximately 20 seconds.



Press  key to remove the message or wait for the screen to timeout.

- The system continues to operate with the configuration of the remaining modules unchanged.
- Communications IDs (e.g. Modbus) of the remaining modules are unchanged regardless of their physical position in the [SenNet Modular](#) system.
- Removed modules may return invalid data on the communications (e.g. Modbus) system.
- Logged data for removed modules is not available.

If a module is removed permanently it is advised that the system is reconfigured using the “*Renumber Slaves*” the option in the *Setup - Advanced* menu. This removes all trace of the modules and rennumbers the other modules to fill the gap.

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Modular Metering System - Configuration

5.3.3. After Replacing Modules

If a module is replaced by an unconfigured module of the same type (for example when replacing a faulty module) the user is given the choice of configuring the replacement module with the same user settings as the module being removed.



The detected module(s) are recognised and shown in their position and the user is requested to accept the changes.

Use the   keys to select YES or NO and the  key to confirm the selection.

NOTE: The “Configuration Changed” screen times out after approximately 20 seconds.

5.3.3.1. Selecting NO (default)

If **NO** is selected (or the screen is allowed to timeout with **YES** or **NO** selected):

- The system continues to operate unchanged with previously configured and accepted modules.
- Replacement modules are skipped when selecting loads in normal operation.
- Existing communications IDs (e.g. Modbus) remain unchanged regardless of physical position in the **SenNet Modular** system.
- New modules are not available on the communications (e.g. Modbus) system.
- Removed modules are not available on the communications (e.g. Modbus) system.
- Logged data for removed modules is not available.

5.3.3.2. Selecting YES

If **YES** is selected.

- New modules are added to the **SenNet Modular** system, in place of removed modules and are made available for display.
- New modules which replace removed modules are automatically configured with the settings of the removed modules.
- Existing communication IDs (e.g. Modbus) remain unchanged providing the number of modules in the **SenNet Modular** system is not changed.
- Logged data for replaced modules is not lost.

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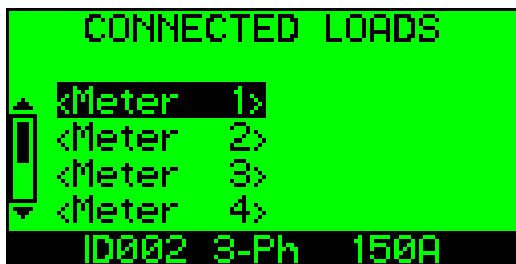
Modular Metering System - Configuration

5.4. *Powering-Up a Configured SenNet Modular System*

- Switch on the auxiliary supply to the unit

The power up screen details the software version for the Master Display Unit and a progress bar shows connection of each Sub Module as it is automatically identified. This screen is displayed for approximately 10 seconds.

- **Connected Loads List**



Use the ▼ ▲ keys to select a load from the list and press  to show the default measurement page for the highlighted load.

The bottom line (Highlighted Text) gives details of the selected module:

- **The Communications ID (eg Modbus ID).**
- **The selected channel (“Ph1”-“Ph3” or “3-Ph” for 3-Phase Loads)**
- **The Current Transducer nominal primary current.**

Un-commissioned systems will display default names and all meters will be set to measure 3-Phase loads. It may be useful to refer back to this **Connected Loads List** after commissioning to check the system configuration.

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Modular Metering System - Configuration

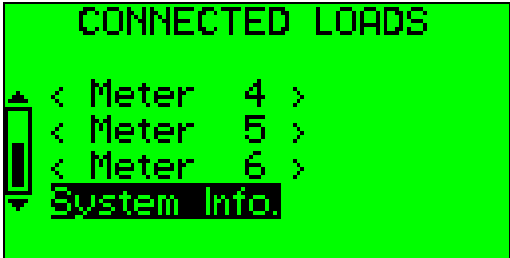
5.4.1. System Information Pages

The system information page provides the user with details of the connected modules at any time during operation of the SenNet Modular system.

Select System Info
From the
CONNECTED LOADS



View the
System Information page



Select to View
Master/Module(s) Information



Return to Module List



Module present and configured



Module not present



Module present but not yet configured

5.4.1.1. Master Information



User Name for the Master Display Unit
Serial Number for the Master Display Unit

Run time (since last power up)

Module status by position

5.4.1.2. Selected Slave Information (MC352)



Selected Module: Serial Number Firmware Version
Voltage: System, Nominal, Calibration Adjust
Current Transformer: Primary Secondary Phase Multiplier
Current Demand Periods (seconds/10)
Power Demand Periods (seconds/10)

Module Status – Selected Module Highlighted

Module Position (Type) Comms ID Meter Type

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- Default Measurement Page



This page shows measured values from the selected meter/load. This data is meaningless for an unconfigured SenNet Modular system as the programmed Current Transducers may not match the physical devices fitted.

The bottom line (Highlighted Text) shows details of the selected load as:

- The Communications ID (eg Modbus ID).
- The selected channel ("Ph1"- "Ph3" or "3-Ph" for 3-Phase Loads)
- The programmed Load Name (Up to 14 Characters)

An LED is illuminated on the selected Module to indicate which phase is displayed on the LCD.

Display Current parameters for the selected load.



Display Voltage parameters for the selected load.



Display Power parameters for the selected load.



Display Energy parameters for the selected load.



Select a load to be displayed (The LED associated with the selected load is illuminated)



Press/Release to show next page
Press/Hold for fast scroll

Press/Release to select next Load
Press/Hold to select from a Load List

Note: A different set of parameter display pages is available for single phase and 3-phase loads. For a list of available pages refer to Section 6.3.

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- **Password Entry Screens**

Some programming features may be protected by a user password. Two levels of password are supported allowing more advanced features to be protected for use only by authorised users.

Move to:

Next / Previous Digit
ESC / OK



Increment / Decrement Digit



Confirm:

OK – Issue the Password
ESC – Return to Prev Screen



NOTE: If a password of 0000 is set, the password entry screen is not shown and immediate access is granted.

If an incorrect password is entered the user is returned to the previous screen.

The factory Set Default Passwords are:

Level 1: 0 0 0 0

Level 2: 1 1 1 1

To change the user passwords refer to Section **XXXX**

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5.5. System Name

Each SenNet Modular system can be given a user defined '*Name*' so it may be individually identified. Names are up to 14 Alpha-Numeric characters and are entered in the *Setup Main* page as follows:

Enter Setup Mode

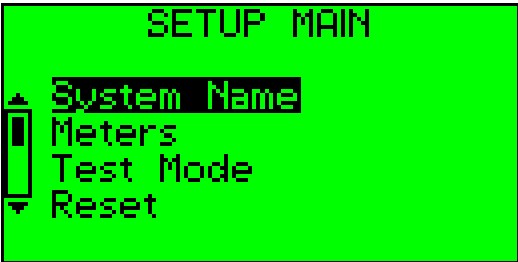


Hold Setup Keys for 5 Seconds

Select System Name from the
SETUP MAIN Menu



View/Edit the System Name

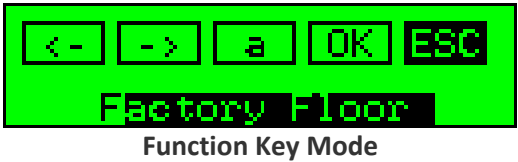
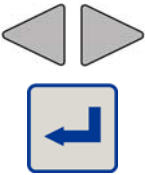


Switch between
Character Entry Mode
and
Function Key Mode



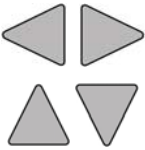
Function Key Mode

- < - Previous Character
- > Next Character
- a/A Lower/Upper Case
- OK Accept New Name
- ESC Abandon Changes

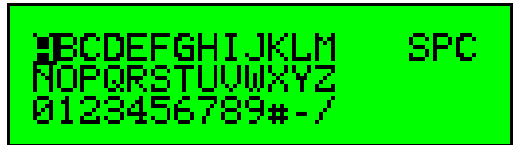


Character Entry Mode

Select Character



Enter Selected Character in Current
position.



Character Entry Mode

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5.6. Setup Each Electricity Meter Module (MC352)

Each Electricity Metering Module **MC352** contains two complete 3-phase electricity meters, each of which may be optionally configured to monitor 3 single-phase loads.

To configure a Meter Module:

- Select the Current Transducer Type matching those connected. Suitable CTs are supplied by the manufacturer and are labelled with the type.
- Configure the meter to measure a 3-phase load or 3 individual single-phase loads.
- Set Autorotate Mode On/Off.
- Give each load a useful name, to be displayed on the LCD and available via Comms.

NOTE: The 3 inputs of each 1/2 MC352 must be connected using the same type of CT.

5.6.1. Select a Current Transducer Type

Enter Setup Mode

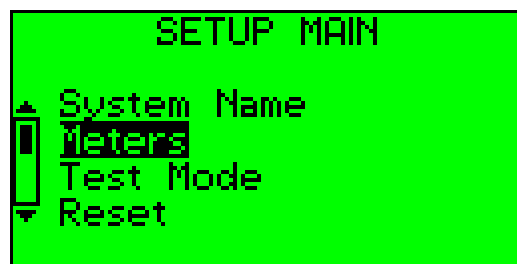


Hold Setup Keys for 5 Seconds

Select Meters
From the Main Menu



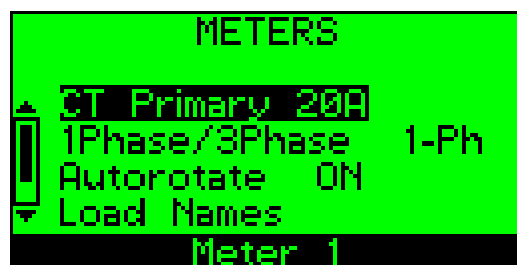
View/Edit the Meter Setup



Select a Meter Module



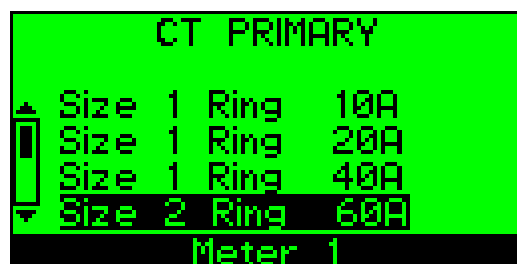
Show CT List



Match the CTs Connected to this
Meter to a type in the List



Accept the CT Selection



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5.6.1.1. If User Spec CT is Selected

It is possible to enter a user CT Specification for each Meter in place of selecting standard devices from the list. This is an advanced option, which is only required in exceptional circumstances and should not be used without technical support. A **Level 2 Password** is required when selecting a **User Spec CT**. This can be entered as described above.

NOTE: The MC352 is designed to compensate for small phase errors produced by each standard current transducer on the list. When a **User Spec CT** is used with the MC352 this phase error should match the specification of the device used or potential measurement errors will occur at low power factors. Current Transducers/Transformers not approved for use with the MC352 may permanently damage the Meter inputs.

5.6.1.2. Enter a User Defined CT Specification

Select **User Spec CT**
From the **CT PRIMARY** list



View/Edit the User Spec CT



```
CT PRIMARY
Split 1      150A
Split 2      400A
Split 3      800A
User Spec CT 20A
Meter 1
```

Select The User Primary



Edit the Primary Current



```
USER CT
User Primary 40A
Phase Offset 0.0
Back
Meter 1
```

Move to:
Next / Previous Digit
ESC / OK



Increment / Decrement Digit



Confirm:
OK – Issue the Password
ESC – Return to Prev Screen



```
ENTER USER CT PRIMARY

00040 [OK] [ESC]
```

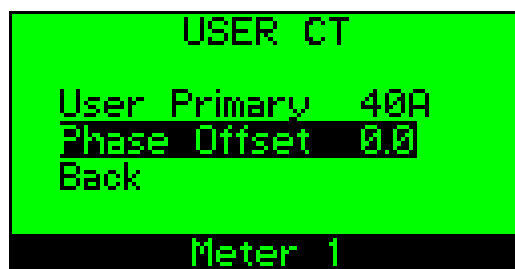

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Modular Metering System - Configuration

Select Phase Offset



Edit the Primary Current



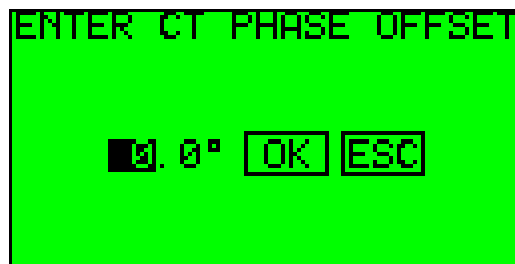
Move to:
Next / Previous Digit
ESC / OK



Increment / Decrement Digit



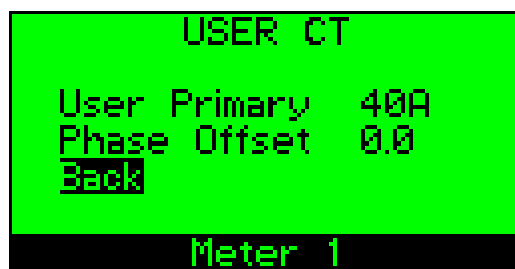
Confirm:
OK – Issue the Password
ESC – Return to Prev Screen



Select Back when User CT is defined.



Accept the User CT Specification
for the selected Meter.



5.6.2. Custom CT Selection

A single **Custom Current Transducer Specification** may be defined, by an authorised user, and will then be available for selection from the Standard CT list.

A **Custom Current Transducer Specification** is defined by setting its CT Primary, and Phase Offset.

NOTE: Current Transducers/Transformers not approved for use with the MC352 may permanently damage the Meter inputs.

NOTE: The Custom CT specification is entered in the Advanced Settings Menu, which requires a Level 2 Password as, described above.

SenNet Modular

Modular Metering System - Configuration

5.6.3. Set the Meter Module to Measure a 3-Phase or 3 Single Phase Loads

Select 1Phase/3Phase

The current mode is shown:

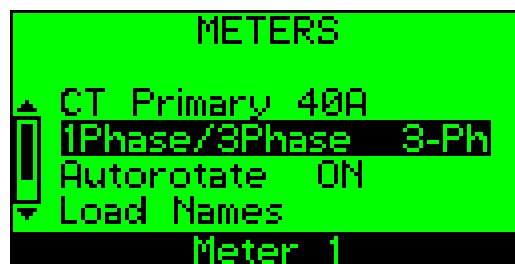
3-Ph or 1-Ph



Toggle Meter Types

3-Ph – 3 Phase Load

1-Ph – 3 x Single Phase



5.6.4. Set Autorotate Mode ON/OFF

When the **Autorotate** function is **ON** (default) the orientation of each current transducer on its associated conductor is irrelevant. If a CT is placed the wrong way round then the **Autorotate** system will automatically compensate for this.

If measurement of **Export power/energy** is required to then it is necessary to switch **Autorotate OFF** and ensure the CTs are mounted the correct way round on the conductors.

Select **Autorotate**

The current mode is shown:

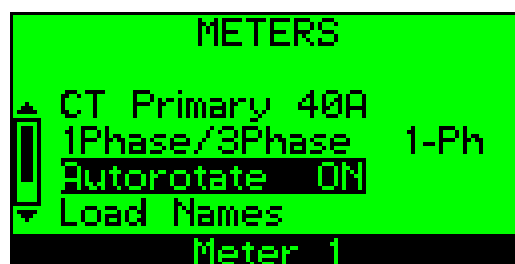
ON or OFF



Toggle **Autorotate**

ON - Normal Operation

OFF – Import/Export Power



SenNet Modular

Modular Metering System - Configuration

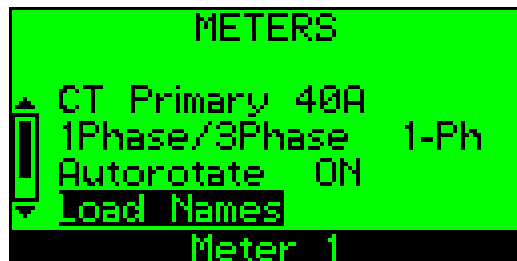
5.6.5. Set a Name For Each Meter

An alphanumeric name may be associated with each load so that it may be identified on the LCD and over the Modbus or other communications link. Each measured load requires a name, so a meter measuring a single 3-phase load requires a single name while a 3 x Single Phase meter requires three names.

Select Load Names



View/Edit the Load Name(s)



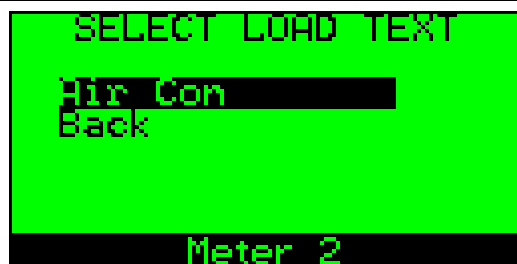
Select The Meter to Edit



Note:

3-Phase Meters Require 1 Name.

1-Phase Meters Require 3 Names

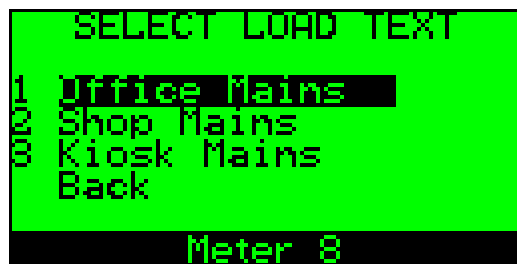


3-Phase Load (Single Name)

Select a Load Name



Edit the Load Name



3 x Single Phase Loads (3 Names)

SenNet Modular

Modular Metering System - Configuration

Switch Between:
Character Entry Mode
and
Function Key Mode



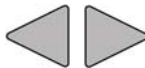
ABCDEFGHIJKLM SPC
NOPQRSTUVWXYZ
0123456789#-/_

<- -> a OK ESC

Office Mains

Function Key Mode

< - Previous Character
- > Next Character
a/A Lower/Upper Case
OK Accept New Name
ESC Abandon Changes



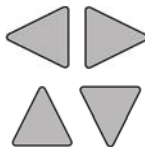
<- -> a OK ESC

Office Mains

Function Key Mode

Character Entry Mode

Select Character



Enter Selected Character in
Current position.



ABCDEFGHIJKLM SPC
NOPQRSTUVWXYZ
0123456789#-/_

Character Entry Mode

5.6.6. Repeat Meter Setup for All Meter Modules

Repeat the programming sequence for all attached loads as described above.

Select **Back** when Meter
Programming is complete



Return to **SETUP** Menu



METERS

1Phase/3Phase 1-Ph
Autorotate ON
Load Names
Back

Meter 8

SenNet Modular

Modular Metering System - Configuration

5.7. Test Mode

A series of load tests may be quickly performed to check the installation and validate connections. To test a load:

- Ensure the load to be tested is ON and in a normal operating mode.
- For import/export meter types ensure the load is importing current
- Step through the load tests
- Isolate the system and correct installation errors where required.

NOTE: Very low Power Factor measurements (<0.55) are used to indicate installation errors. If the load measured actually produces such a low PF then test results may be invalid.

NOTE: If more than one error occurs simultaneously the first error in the list below will be shown until the error has been corrected.

5.7.1. Installation Errors:



No Error Detected

- The Current Transformer appears to be on the correct phase (Power Factor > 0.55).
- Sufficient load current is flowing for a valid test.
- The phase voltage is healthy.



Insufficient Phase Voltage to carry out a Valid Test

- Check that the voltage wiring and ensure circuit breakers are ok.



Insufficient Load Current to carry out a Valid Test

- Check that the load is running before carrying out the test.



Current Transformer Mounted on the Wrong Phase Cable

- This results in low power factor and kW measurements.
- Check the wiring and move the CTs to the correct phase.



Current Transformer Reversed on its Cable (Auto CT reverse activated)

- This occurs when the CT is reversed on its cable and indicates that the meter is automatically correcting for this error.
- There is no need to rotate the CT in this case this symbol is shown purely for information.



Current Transformer Reversed on its Cable (Import/export meter)

- This result is only valid for import/export meter types.
- This results in negative kW measurements.
- Check the wiring and rotate the CT on its phase cable.

SenNet Modular

Modular Metering System - Configuration

Enter Setup Mode

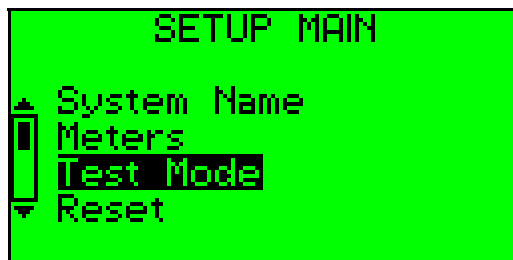


Hold Setup Keys for 5 Seconds

Select Test Mode from the Main Menu



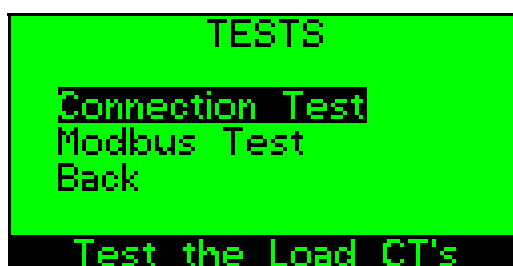
Enter Test Mode



Select Connection Test



Enter Connection Test Mode



Connection Test - Power Factor

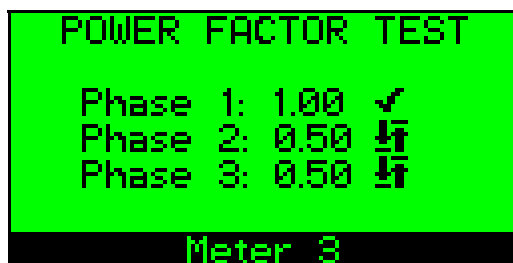
Select Meter to Test



Next / Previous Test



Exit Connection Test Mode



Connection Test – Amps

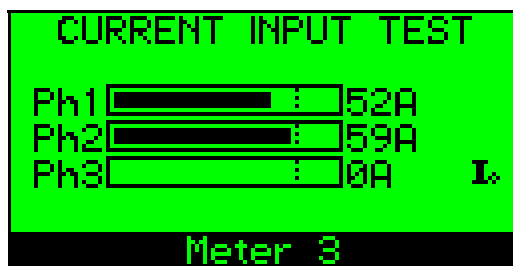
Select Meter to Test



Next / Previous Test



Exit Test Mode



SenNet Modular

Modular Metering System - Configuration

Connection Test – Volts

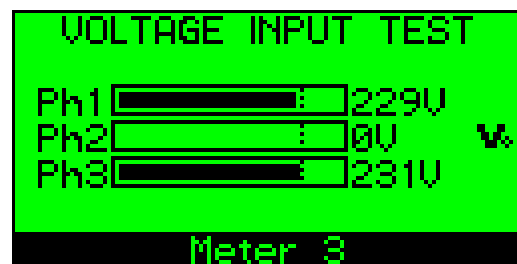
Select Load to Test



Next / Previous Test



Exit Test Mode



Connection Test – CT Rotation

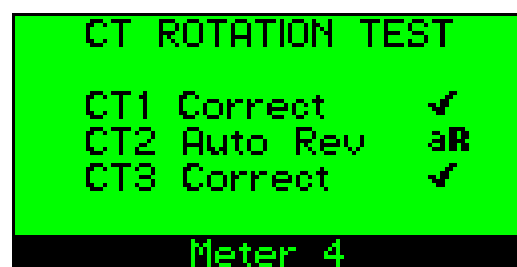
Select Load to Test



Next / Previous Test



Exit Test Mode



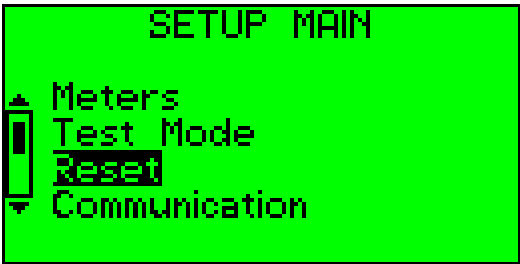
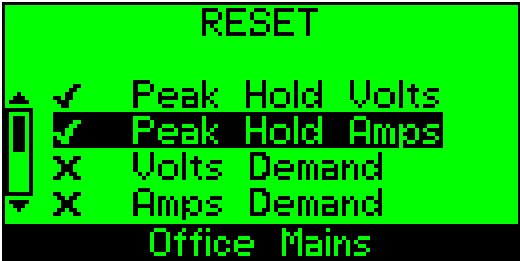
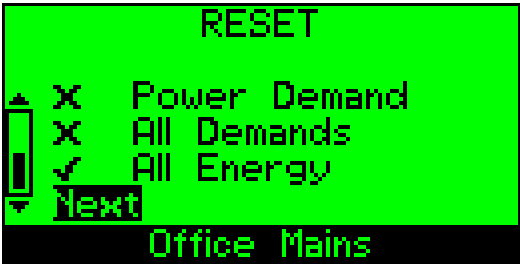
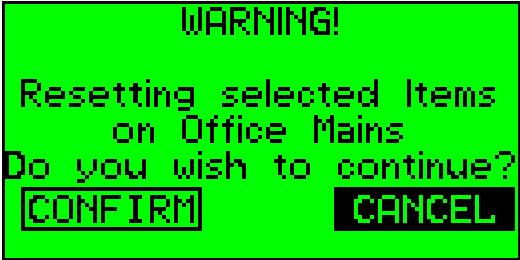
SenNet Modular

Modular Metering System - Configuration

5.8. Reset Saved Data

Various stored parameters may be reset to zero using the programming menus.

- “All Loads” – Reset all meter modules simultaneously
- “All Demands” – Reset Volts, Amps and Power Demands simultaneously
- “All Energy” – Reset all energy registers simultaneously

Enter Setup Mode	 	Hold Setup Keys for 5 Seconds
Select Reset from the <i>SETUP MAIN</i> Menu	 	
Enter the Reset Menu *		
Select a single load or All Loads to reset.		
Select One or More Parameters to Reset	 	
Toggle Between:		
✓ Reset Stored Value ✗ Don't Reset Stored Value		
Select Next from the <i>RESET</i> Menu	 	
Proceed With Reset **		
<u>WARNING MESSAGE</u>		
CONFIRM: Perform Resets CANCEL: Exit Without Reset		

* If passwords are enabled enter your password as described above.

** Exit to main menu if no parameters are selected

SenNet Modular

Modular Metering System - Configuration

5.9. Setup Modbus RS485 Communications

Each **SenNet Modular** fitted with the Modbus RS485 communications

The user assigns a unique ID for each **SenNet Modular** in the Modbus network leaving gaps for the Master Display Unit to automatically assign a further set of consecutive IDs (2 for each module).

- The user selects a baud rate to suit the Modbus system setting.
- The user set the Parity to suit the Modbus system setting.

Note: Parity is set to NONE for the majority of Modbus systems.

Enter The *SETUP MAIN* Menu

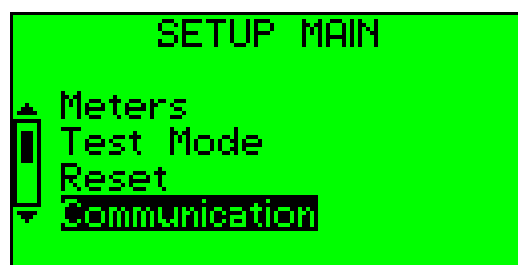


Hold Setup Keys for 5 Seconds

Select *Communication* from the *SETUP MAIN* menu



Enter the Communication Setup Menu

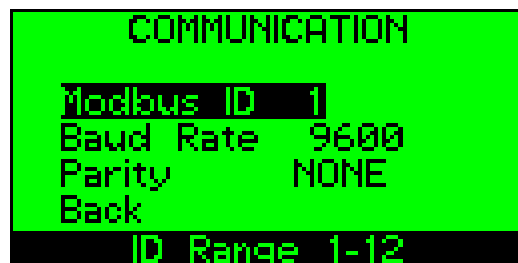


Set The Modbus ID

Select Modbus ID from the menu



Enter to change the value



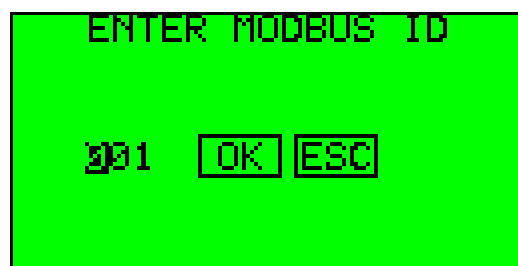
Select a *Digit*, *ESC* or *OK*



Change the Digit Value.



OK – Accept new value
ESC- Reject new value



SenNet Modular

Modular Metering System - Configuration

Set The Baud Rate

Select *Baud Rate* from the
Communication Menu



Enter to Change the Value



```
COMMUNICATION
Modbus ID  1
Baud Rate  9600
Parity      NONE
Back
9600-NO-1
```

Select the desired baud rate



Enter to accept the selection



```
BAUD RATE
4800
9600
14400
19200
Meter 1
```

Set The Parity

Select *Parity* from the menu.



Enter to Change the Value



```
COMMUNICATION
Modbus ID  1
Baud Rate  9600
Parity      NONE
Back
9600-NO-1
```

Return To the *SETUP MAIN* Menu

Select *Back* from the menu



Exit to *SETUP MAIN* menu



```
COMMUNICATION
Modbus ID  1
Baud Rate  9600
Parity      NONE
Back
9600-NO-1
```

SenNet Modular

Modular Metering System - Operation

6. Meter Operation

Different measured loads are shown on the graphics LCD using the keypad on the **Master Display Unit**. Different sets of display pages are available for 1-Phase and 3-Phase loads.

6.1. Display Scaling

6.1.1. Voltage Scaling

The voltage display normally shows a 4 digit number scaled to 1 decimal place:

Examples: 230.0V, 400.0V

6.1.2. Current Scaling

The display resolution for all current (amps) parameters is set to provide 4 digits of resolution at the nominal CT Primary input.

<i>CT Primary (Amps)</i>	<i>Current Display Resolution</i>
1A to 8A	0.001 A
9A to 80A	0.01 A
81A to 800A	0.1A
801A to 8000A	1 A
> 8000A	0.01 kA

6.1.3. Power Scaling (Meters Configured for a Single 3-Phase Load)

<i>CT Primary (Amps)</i>	<i>3-Phase Power Display Resolution</i>
< 5A	0.001kW
5A – 35A	0.01kW
36A to 350A	0.1kW
351A to 3500A	1kW
>3500A	0.01MW

6.1.4. Power Scaling (Meters Configured for a 3 x Single Phase Loads)

<i>CT Primary (Amps)</i>	<i>1 Phase Power Display Resolution</i>
<11A	0.001kW
11A to 100A	0.01kW
101A to 1000A	0.1kW
1001A to 10,000A	1kW
> 10,000A	0.01MW

6.1.5. Energy Scaling (Meters Configured for a Single 3-Phase Load)

<i>CT Primary (Amps)</i>	<i>3 Phase Energy Display Resolution</i>
< 5A	0.001kW
5A – 35A	0.01kW
36A to 350A	0.1kW
351A to 3500A	1kW
>3500A	0.01MW

SenNet Modular

Modular Metering System - Operation

6.1.6. Energy Scaling (Meters Configured for 3 x Single Phase Loads)

<i>CT Primary (Amps)</i>	<i>1 Phase Energy Display Resolution</i>
<11A	0.001kW
11A to 100A	0.01kW
101A to 1000A	0.1kW
1001A to 10,000A	1kW
> 10,000A	0.01MW

6.2. Sliding Window Demand

Average values of Volts, Amps kW, and kvar are calculated over a user programmable time period. The displays show the averages for the most recent time period ending at the time the display was last updated. The demand period is continuously updated as time progresses hence the term “**Sliding Window Demand**”.

SenNet Modular

Modular Metering System - Operation

6.3. Load Display Menus

Each metered load is represented in display menus which are accessed using the user keypad on the *Master Display Unit*.

6.3.1. Capacitive and Inductive Loads

Measured parameters such as kvar and Power Factor are displayed with a symbol indicating the type of load:

Inductive Loads: 

Capacitive Loads: 

6.3.2. 3-Phase Load Display Menus

3-Phase Current Display Menu



Select Current Pages

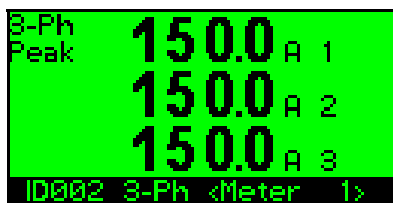


Select Load



Instantaneous Phase Currents

Phase 1 Current
Phase 2 Current
Phase 3 Current



Peak Hold Phase Currents

Peak Hold Phase 1 Current
Peak Hold Phase 2 Current
Peak Hold Phase 3 Current



Current Demand (Sliding Window) Ref 6.2

Current Demand Phase 1
Current Demand Phase 2
Current Demand Phase 3



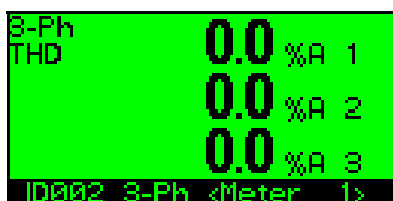
Peak Current Demand (Sliding Window) Ref 6.1

Peak Hold Current Demand Phase 1
Peak Hold Current Demand Phase 2
Peak Hold Current Demand Phase 3



Min Current Demand (Sliding Window) Ref 6.1

Minimum Hold Current Demand Phase 1
Minimum Hold Current Demand Phase 2
Minimum Hold Current Demand Phase 3



Current Total Harmonic Distortion (THD)

%THD Phase 1 Current
%THD Phase 2 Current
%THD Phase 3 Current

SenNet Modular

Modular Metering System - Operation

3-Phase Voltage Display Menu



Select Voltage Pages



Select Load



Instantaneous Phase to Neutral Voltages

Phase 1 to neutral volts
Phase 2 to neutral volts
Phase 3 to neutral volts



Instantaneous Line-Line Voltages

Line 1 – Line 2 volts
Line 2 – Line 3 volts
Line 3 – Line 1 volts



Peak Hold Phase to Neutral Voltages

Peak Hold Phase 1 Volts
Peak Hold Phase 2 Volts
Peak Hold Phase 3 Volts



Voltage Demand (Sliding Window) Ref 6.1

Voltage Demand Phase 1
Voltage Demand Phase 2
Voltage Demand Phase 3



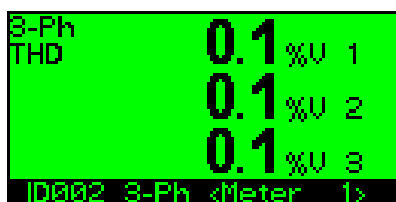
Peak Voltage Demand (Sliding Window) Ref 6.1

Peak Hold Voltage Demand Phase 1
Peak Hold Voltage Demand Phase 2
Peak Hold Voltage Demand Phase 3



Min Voltage Demand (Sliding Window) Ref 6.1

Minimum Hold Voltage Demand Phase 1
Minimum Hold Voltage Demand Phase 2
Minimum Hold Voltage Demand Phase 3



Voltage Total Harmonic

%THD Phase 1 Volts
%THD Phase 2 Volts
%THD Phase 3 Volts

SenNet Modular

Modular Metering System - Operation

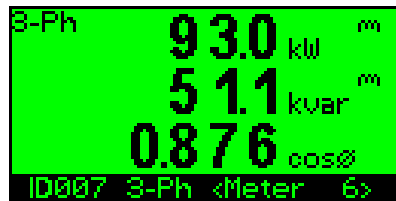
3-Phase Power Display Menu



Select Power Pages



Select Load



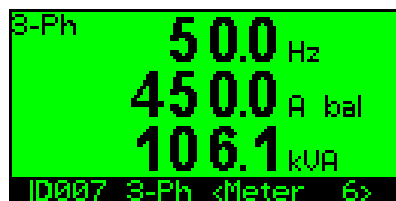
Instantaneous System Power

System Real Power kW (P)
System Reactive kvar (Q)
System Power Factor (COSΦ)



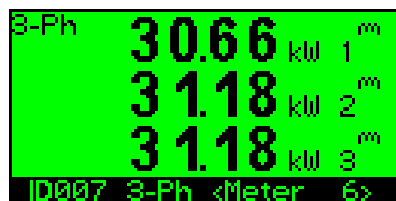
kW Demand (Sliding Window) Ref 6.1

kW Demand
Peak Hold kW Demand
Minimum Hold kW Demand



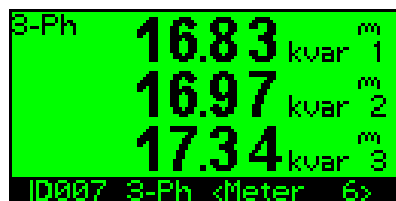
System Frequency, Neutral Current, kVA (S)

Frequency (Measured on Volts Ph1)
Neutral Current Ref ¡Error! No se encuentra el origen de la referencia.
System Apparent Power kVA



Per Phase kW (P1-P3)

Phase 1 Real Power (kW)
Phase 2 Real Power (kW)
Phase 3 Real Power (kW)



Per Phase Reactive Power (kvar)

Phase 1 Reactive Power (Inductive shown)
Phase 2 Reactive Power (Inductive shown)
Phase 3 Reactive Power (Inductive shown)



Per Phase Power Factor (COS Ø)

Phase 1 Power Factor (Capacitive shown)
Phase 2 Power Factor (Capacitive shown)
Phase 3 Power Factor (Capacitive shown)

SenNet Modular

Modular Metering System - Operation

3-Phase Energy Display Menu



Select Energy Pages



Select Load



Total System Import Energy

Real Energy kWh

Reactive Energy (kvarh)

Apparent Energy (kVAh)r

SenNet Modular

Modular Metering System - Operation

6.3.3. Single-Phase Meter Display Menus

NOTE: Each single-phase load is associated with a phase voltage determined by its position in a 3-Phase metering module. The phase voltages connected to the Master Display Unit are numbered Ph1 - Ph3 and this is indicated on the single-phase pages as "Ph1" - "Ph3".

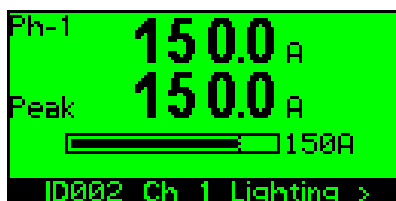
Single-Phase Current Display Menu



Select Current Pages



Select Load



Instantaneous Phase Current

Phase Current

Peak Hold Phase Current

Bar Graph of Amps Scale = 0 - 120% CT Prim



Current Demand (Sliding Window) Ref 6.1

Phase Current Demand

Peak Hold Phase Current Demand

Minimum Hold Phase Voltage Demand

Single-Phase Voltage Display Menu



Select Voltage Pages



Select Load



Instantaneous Phase to Neutral Voltage

Phase to Neutral Voltage

Peak Hold Phase Voltage

Bar Graph of Volts Scale = 0 - 120% Vnom



Voltage Demand (Sliding Window) Ref 6.1

Phase Voltage Demand

Peak Hold Phase Voltage Demand

Minimum Hold Phase Voltage Demand

SenNet Modular

Modular Metering System - Operation

Single-Phase Power Display Menu



Select Power Pages



Select Load



Instantaneous Phase Powers

- Single Phase Real Power (P)
- Single Phase Reactive Power (Q)
- Single Phase Power Factor (COSΦ)



Power Demand (Sliding Window) Ref 6.1

- Single Phase kW Demand
- Single Phase Peak Hold kW Demand
- Frequency (From Phase 1 volts)

Single-Phase Energy Display Menu



Select Energy Pages



Select Load



Single Phase Total Import Energy

- Single Phase Real Energy (kWh)
- Single Phase Reactive Energy (kvarh)

SenNet Modular

Modular Metering System - Specification

7. Specification

7.1. Modular Metering System

ENVIRONMENT		
Temperature	Operating	-10°C to +55°C
	Storage	-25°C to +70°C
Humidity	< 75% non-condensing	
Altitude	<= 2000m	
Environment	IP54 standard	
MECHANICAL		
Enclosure	Material: Black ABS UL 94-V0	
Dimensions	Height :	164mm
	Depth: (Off Wall)	96mm
	Length:	
	Master Display Unit:	100mm
	Communication Module:	29mm
	Dual Metering Slave:	29mm
SAFETY		
Conforms to	EN 61010-1 Overvoltage Category III, Pollution Degree 2	

SenNet Modular

Modular Metering System - Specification

7.2. Master Display Module

GENERAL		
Dimensions	Height :	164mm
	Depth: (Off Wall)	96mm
	Length:	100mm
System	1-Phase 2 Wire Unbalanced Loads 3-Phase 3 or 4 Wire Unbalanced Loads 2 Phase 3 wire Unbalanced Loads	
Modules (Capacity)	Maximum = 10 Modules e.g. 10x Metering Modules MC352	
Voltage Un	Nominal:	400V Phase to Phase (Optional 480V Phase-Phase) 230V. Phase to Vn (Optional 277Vac Phase-Neutra))
	Terminals Rising Cage. 2.5mm2 (12 AWG) cable max.	
Current In	Nominal:	0.33V from Externally Isolated Custom Current Transducers.
Measurement Range	Voltage	40% to 120% of Nominal
	Current	0.2% to 120% of Nominal
Frequency Range	Fundamental	45 to 65Hz
	Harmonics	Up to 30th harmonic at 50Hz Individual to the 15th
Burden	Voltage	<0.1VA per phase
	Current	Not applicable (Burden fitted in transducer)
Overload	Voltage	x4 for 1 hour
	Current	x2 permanently
DISPLAY		
Type	LCD 128x64 Dot Graphic	
AUXILIARY SUPPLY		
Input Range	100-240Vac ±15% 45-65Hz	
Load	Master Only :	3 W max
	Master + 10 MC352 Modules:	5 W max.

SenNet Modular

Modular Metering System - Specification

7.3. Dual Metering Module MC352

GENERAL		
Dimensions	Height :	164mm
	Depth: (Off Wall)	96mm
	Length:	29mm
Current Inputs: (1/2 Module)	Use only Northern Design Current Transducers with the following specification:	
	Nominal CT Primary: 20A;40A;60A;100A; 150A; 200A, 300A, 400A; 600A, 800A, 1000A or 1200A	
	Nominal CT Secondary: 0.333Vac	
	Insulation Class 600Vac	
Voltage Input	From Master Display Unit: maximum 0V - 3.3V ac Peak	
ACCURACY All errors ± 1 digit		
kWh	Better than Class 1 per EN 62053-21 & BS 8431	
kvarh	Better than Class 2 per EN 62053-23 & BS 8431	
kW & kVA	Better than Class 0.25 IEC 60688	
kvar	Better than Class 0.5 IEC 60688	
Amps & Volts	Class 0.1 IEC 60688 (0.01In – 1.2In or 0.1Un – 1.2Un)	
PF	±0.2° (0.05In – 1.2In and 0.2Un – 1.2Un)	
Neutral Current	Class 0.5 IEC 60688 (0.05In – 1.2In)	
POWER SUPPLY		
DC Power From Master Display	DC Power Supply:	3.3V DC
	Maximum Load Per Module (2 Meters) : 0.15W	